

ECOLE POLYTECHNIQUE

ESPCI

2026 ADMISSION COMPETITION

TUESDAY, APRIL 14, 2026

08:00 - 12:00

MP-MPI TRACKS - Paper No. 3

MATHEMATICS B

Duration: 4 hours

The use of calculators is not permitted for this paper.

English translation

MATHEMATICS PAPER

(Duration: 4 hours)

The use of calculators is not permitted for this paper.

Notation

If n is a positive integer, let $\mathcal{M}_n(\mathbb{R})$ denote the set of matrices with n rows and n columns and real coefficients.

If $M \in \mathcal{M}_n(\mathbb{R})$ and if i and j are two integers between 1 and n , $M_{i,j}$ denotes the coefficient of M located in row i and column j .

If $M \in \mathcal{M}_n(\mathbb{R})$, $\text{Sp}(M)$ denotes the real spectrum of the matrix M , that is, the set of its real eigenvalues with their multiplicities. More precisely, $\text{Sp}(M)$ is the subset of $\mathbb{R} \times \mathbb{N}$ whose elements are the pairs (λ, m_λ) where λ is a real eigenvalue of M and m_λ is its multiplicity as a root of the characteristic polynomial of M . We also denote by $\text{Tr}(M)$ the trace of M .

If i and j are two relative integers, $\delta_{i,j}$ denotes the integer 1 if $i = j$ and 0 if $i \neq j$.

If A is a subset of $\mathbb{R}$, $\mathbf{1}_A$ denotes the function from $\mathbb{R}$ to $\mathbb{R}$ that is equal to 1 on A and to 0 on the complement of A .

Throughout the problem, $(\Omega, \mathcal{A}, \mathbb{P})$ denotes a probability space on which the various random variables in the problem will be defined. We shall admit that all random variables introduced can indeed be constructed on this space. We write $\mathbb{P}(A)$ for the probability of an event $A \in \mathcal{A}$ and $\mathbb{E}(Z)$ for the expectation of a real-valued random variable Z on $(\Omega, \mathcal{A}, \mathbb{P})$.

In this paper, to solve a question, one may use the results of the preceding questions. Part III is independent of Parts I and II, and Part IV is essentially independent of Parts I and II.

Preliminary

1. Let $\alpha \in \mathbb{R}$. Consider the real sequence $(u_n)_{n \geq 0}$ defined by

$$u_0 = 0, \quad u_1 = 1, \quad \forall n \geq 2, \quad u_n = \alpha u_{n-1} - u_{n-2}.$$

Express, for every integer $n \geq 0$, the term u_n in terms of n and α . Carefully distinguish the following cases: $|\alpha| > 2$, $|\alpha| < 2$, $\alpha = 2$, and $\alpha = -2$.

Part I

2. Let f be a real-valued function defined and continuous on $[0, 1]$. Consider the real sequences $(v_n)_{n \geq 1}$ and $(w_n)_{n \geq 1}$ defined by

$$\forall n \geq 1, \quad v_n = \frac{1}{n} \sum_{k=1}^n f\left(\frac{k}{n+1}\right), \quad w_n = \frac{1}{n} \sum_{k=1}^n f\left(\frac{2k}{2n+1}\right).$$

Show that the sequences $(v_n)_{n \geq 1}$ and $(w_n)_{n \geq 1}$ converge and have the same limit.

3a. Let f be a real-valued function defined and continuous on $[-2, 2]$. Prove that the integral

$$I(f) = \frac{1}{\pi} \int_{-2}^2 \frac{f(x)}{\sqrt{4-x^2}} dx$$

is convergent, and that

$$I(f) = \frac{1}{\pi} \int_0^\pi f(2 \cos \theta) d\theta.$$

3b. For a natural integer n , let f_n be the function from $\mathbb{R}$ to $\mathbb{R}$ defined by $f_n(x) = x^n$. Compute the value of the integral $I(f_n)$ when n is equal to 0, 1, and 2.

3c. For every natural integer $n \geq 0$, compute the value of the integral $I(f_n)$ in terms of n .

In the following questions, for every integer $n \geq 1$, let U_n be a discrete uniform random variable with values in the natural integers between 1 and n , that is,

$$\forall k \in \{1, \dots, n\}, \quad \mathbb{P}(U_n = k) = \frac{1}{n}.$$

4a. If f is a real-valued function defined and continuous on $[-2, 2]$, prove that

$$\mathbb{E} \left(f \left(2 \cos \left(\frac{\pi U_n}{n+1} \right) \right) \right) \xrightarrow{n \rightarrow \infty} \frac{1}{\pi} \int_{-2}^2 \frac{f(x)}{\sqrt{4-x^2}} dx.$$

4b. Prove that, for every $y \in [-2, 2]$,

$$\mathbb{P} \left(2 \cos \left(\frac{\pi U_n}{n+1} \right) < y \right) \xrightarrow{n \rightarrow \infty} \frac{1}{\pi} \int_{-2}^y \frac{1}{\sqrt{4-x^2}} dx.$$

Part II

Let $n \geq 2$ be an integer. Let T_n be the matrix defined by

$$(T_n)_{j,k} = \delta_{j,k+1} + \delta_{j+1,k}, \quad 1 \leq j, k \leq n.$$

In other words,

$$T_n = \begin{pmatrix} 0 & 1 & 0 & \cdots & 0 \\ 1 & 0 & 1 & \ddots & \vdots \\ 0 & 1 & 0 & \ddots & 0 \\ \vdots & \ddots & \ddots & \ddots & 1 \\ 0 & \cdots & 0 & 1 & 0 \end{pmatrix}$$

and $\chi_n(X) = \det(XI_n - T_n)$ denotes the characteristic polynomial of T_n .

5a. For $n = 2$ and $n = 3$, compute the polynomial χ_n and determine the spectrum of T_n .

5b. For every integer $n \geq 4$, express χ_n in terms of χ_{n-1} and χ_{n-2} .

5c. Let α be a complex number such that $|\alpha| < 2$. Establish the following general formula for $\chi_n(\alpha)$, when $n \geq 2$:

$$\chi_n(\alpha) = \frac{1}{i\sqrt{4-\alpha^2}} \left(\left(\frac{\alpha + i\sqrt{4-\alpha^2}}{2} \right)^{n+1} - \left(\frac{\alpha - i\sqrt{4-\alpha^2}}{2} \right)^{n+1} \right).$$

5d. Deduce an exact expression for the coefficients of the polynomial χ_n (the coefficients may be given as a sum of products of binomial coefficients).

6. For an integer $n \geq 2$, show that the eigenvalues of T_n are given by the formula

$$2 \cos \left(\frac{k\pi}{n+1} \right), \quad k = 1, \dots, n.$$

If f is a real-valued function on $\mathbb{R}$ and $M \in \mathcal{M}_n(\mathbb{R})$, define

$$S_f(M) = \frac{1}{n} \sum_{(\lambda, m_\lambda) \in \text{Sp}(M)} m_\lambda f(\lambda).$$

7. Prove that if f is defined and continuous on $[-2, 2]$, then

$$\lim_{n \rightarrow +\infty} S_f(T_n) = \frac{1}{\pi} \int_{-2}^2 \frac{f(x)}{\sqrt{4-x^2}} dx.$$

Let $n \geq 2$ be an integer, and let a, b, c be three real numbers. Let $T_n(a, b, c)$ be the matrix in $\mathcal{M}_n(\mathbb{R})$ such that, for $1 \leq i, j \leq n$,

$$(T_n(a, b, c))_{ij} = a\delta_{i,j} + b\delta_{i+1,j} + c\delta_{i,j+1},$$

that is,

$$T_n(a, b, c) = \begin{pmatrix} a & b & 0 & \cdots & 0 \\ c & a & b & \ddots & \vdots \\ 0 & c & a & \ddots & 0 \\ \vdots & \ddots & \ddots & \ddots & b \\ 0 & \cdots & 0 & c & a \end{pmatrix}.$$

8a. Express the spectrum of $T_n(a, b, c)$ in terms of a and the spectrum of $T_n(0, b, c)$.

8b. Express the spectrum of $T_n(a, b, c)$ in terms of a and the spectrum of $T_n(0, bc, 1)$.

8c. Suppose that $bc > 0$. Express all complex eigenvalues of $T_n(a, b, c)$ in terms of a, b, c , and n .

Hint: one may first reduce to the case $b = c$.

From now on, assume that a, b, c are three real numbers with $bc > 0$.

9a. Prove that for every real-valued continuous function f on $\mathbb{R}$, one has

$$\lim_{n \rightarrow \infty} S_f(T_n(a, b, c)) = \frac{1}{\pi} \int_{-2}^2 \frac{f(a + \sqrt{bc}x)}{\sqrt{4 - x^2}} dx.$$

9b. Let $y \in \mathbb{R}$, and let $q_n(y)$ be the number of eigenvalues of $T_n(a, b, c)$ in the interval $(-\infty, y]$. Give an asymptotic equivalent of $q_n(y)$ as $n \rightarrow \infty$.

Part III

The aim of this part is to give a proof of the Weierstrass approximation theorem. The use of this theorem is therefore not permitted in Questions 10 to 12.

Let $n \geq 0$ be an integer. Let Q_n be the polynomial

$$Q_n = (1 - X^n)^{2^n} \in \mathbb{R}[X]$$

and let P_n be the polynomial $P_n = Q_n((1 - X)/2)$. Also let H be the function from $[-1, 1]$ to $\mathbb{R}$ defined by

$$\forall x \in [-1, 1], \quad H(x) = \begin{cases} 0, & \text{if } x < 0, \\ 1, & \text{if } x \geq 0. \end{cases}$$

10a. Show that, for every real number $0 \leq \kappa < \frac{1}{2}$, the sequence $(Q_n)_{n \geq 0}$ converges uniformly to 1 on the interval $[0, \kappa]$ and to 0 on the interval $[1 - \kappa, 1]$.

10b. Deduce that, for every real number $0 < \eta \leq 1$, the sequence $(P_n)_{n \geq 0}$ converges uniformly to H on $[-1, 1] \setminus [-\eta, \eta]$.

Let f be a continuous function from $[-1, 1]$ to $\mathbb{R}$, and let $\varepsilon > 0$.

11. Assume, in this question only, that $f(-1) = 0$. Show that there exist an integer $N \geq 1$, real numbers

$$-1 < c_1 < c_2 < \cdots < c_N < 1$$

and $(a_1, \dots, a_N) \in [-\varepsilon, \varepsilon]^N$ such that

$$\forall x \in [-1, 1], \quad \left| f(x) - \sum_{i=1}^N a_i H(x - c_i) \right| \leq \varepsilon.$$

12. Deduce that there exists a polynomial $P \in \mathbb{R}[X]$ such that

$$\forall x \in [-1, 1], \quad |f(x) - P(x)| \leq \varepsilon.$$

Hint: one may consider a polynomial of the form

$$F_{\varepsilon, n} = \sum_{i=1}^N a_i P_n(X - c_i)$$

and choose a real number $\eta > 0$ such that the intervals $[c_i - \eta, c_i + \eta]$ are pairwise disjoint.

Part IV

Let $(W_{i,j})_{1 \leq i \leq j}$ be a family of independent identically distributed random variables with values in $\mathbb{Z}$ such that

$$\mathbb{E}(W_{1,1}) = 0, \quad \mathbb{E}(W_{1,1}^2) = 1, \quad |W_{1,1}|^k \text{ has finite expectation for every integer } k \geq 0.$$

For every integer $n \geq 1$ and every $\omega \in \Omega$, let $X_n(\omega) \in \mathcal{M}_n(\mathbb{R})$ be defined by

$$\forall 1 \leq i, j \leq n, \quad (X_n(\omega))_{i,j} = \begin{cases} \frac{W_{i,j}(\omega)}{\sqrt{n}}, & \text{if } i \leq j, \\ \frac{W_{j,i}(\omega)}{\sqrt{n}}, & \text{if } j \leq i. \end{cases}$$

If f is a function from $\mathbb{R}$ to $\mathbb{R}$, let $S_n(f)$ be the random variable defined by

$$\forall \omega \in \Omega, \quad S_n(f)(\omega) = S_f(X_n(\omega)),$$

where the notation $S_f(M)$ for $M \in \mathcal{M}_n(\mathbb{R})$ was defined before Question 7. We also write, when f is continuous,

$$\Sigma(f) = \frac{1}{2\pi} \int_{-2}^2 f(x) \sqrt{4 - x^2} dx.$$

For every natural integer $k \geq 0$, denote by (H_k) the following hypothesis:

$$(H_k) \quad \lim_{n \rightarrow +\infty} \frac{1}{n} \mathbb{E}(\text{Tr}(X_n^k)) = \Sigma(f_k) \quad \text{and} \quad \lim_{n \rightarrow +\infty} \frac{1}{n^2} \mathbb{E}(\text{Tr}(X_n^k)^2) = \Sigma(f_k)^2,$$

where, recall, f_k denotes the function $x \mapsto x^k$ from $\mathbb{R}$ to $\mathbb{R}$.

13a. For every integer $k \geq 0$ and every integer $n \geq 1$, show that one has the following equality of random variables:

$$S_n(f_k) = \frac{1}{n} \text{Tr}(X_n^k).$$

13b. Compute $\Sigma(f_k)$ in terms of k for every integer $k \geq 0$.

13c. Prove (H_k) for $0 \leq k \leq 2$.

In the remainder of the paper, assume from now on that (H_k) is verified for every integer $k \geq 0$.

14. For every integer $k \geq 0$ and every real number $B > 0$, let $g_{k,B}$ be the function defined on $\mathbb{R}$ by

$$\forall x \in \mathbb{R}, \quad g_{k,B}(x) = |x|^k \mathbf{1}_{\{|x| > B\}}.$$

Show that, for $\varepsilon > 0$,

$$\mathbb{P}(S_n(g_{k,B}) \geq \varepsilon) \leq \frac{\mathbb{E}(S_n(f_{2k}))}{\varepsilon B^k}.$$

15. Deduce that for all real numbers $B > 4$ and $\varepsilon > 0$, one has

$$\lim_{n \rightarrow +\infty} \mathbb{P}(S_n(g_{k,B}) \geq \varepsilon) = 0.$$

Hint: one may observe that, for $k \leq k'$ and $B > 4$, one has $g_{k,B} \leq g_{k',B}$.

From now on, fix a real number $B > 4$.

16. Show that for every integer $k \geq 0$ and every real number $\varepsilon > 0$, one has

$$\lim_{n \rightarrow +\infty} \mathbb{P}(|S_n(f_k) - \mathbb{E}(S_n(f_k))| \geq \varepsilon) = 0.$$

Let f be a continuous function from $\mathbb{R}$ to $\mathbb{R}$ that is zero outside $[-B, B]$. Fix a real number $\varepsilon > 0$.

17a. Show that there exist a polynomial $P \in \mathbb{R}[X]$ and an integer $N \geq 0$ such that, for every $n \geq N$,

$$\mathbb{P}(|S_n(f) - \Sigma(f)| \geq \varepsilon) \leq \mathbb{P}(|S_n(P\mathbf{1}_{\{|x|>B\}})| \geq \varepsilon/4) + \mathbb{P}(|S_n(P) - \mathbb{E}(S_n(P))| \geq \varepsilon/4).$$

17b. Conclude that

$$\lim_{n \rightarrow +\infty} \mathbb{P}(|S_n(f) - \Sigma(f)| \geq \varepsilon) = 0.$$